

TUSHAR SHRIKANT RATHOD

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Leetcode-2132 | Codechef-5-Star | Hackerrank

WORK EXPERIENCE

Wayfair | Software Engineer 1 (Data Tooling)

07/25 – Present

- **Architected and took full ownership** of a multi-agent GenAI Data Contract Creation platform end-to-end, translating highly ambiguous dataset requirements into a scalable system that reduced contract creation time from days to minutes.
- **Integrated BigQuery** for real-time relational database schema discovery, metadata inference, and live type validation, ensuring high-quality, version-controlled data governance across distributed teams.
- **Engineered robust distributed data pipelines** using Apache Airflow DAGs, defining strict production testing standards, managing complex workflow orchestration, and independently troubleshooting production scheduling systems.

Wayfair | Software Engineer Intern

01/25 – 06/25

- **Developed a scalable repository analysis system** using an advanced Retrieval-Augmented Generation (RAG) architecture to automate the discovery of Apache Kafka usage patterns across a massive, heterogeneous codebase.
- **Optimized distributed data retrieval** by implementing intelligent file chunking and generating vector database embeddings, drastically reducing the manual engineering effort required for an enterprise-wide Confluent Kafka migration.

Catalinko | Backend Intern

03/24 – 05/24

- **Designed and architected scalable backend services** for document storage, inventory management, order processing, and payment integration using AWS Lambda, AWS AppSync, and AWS CloudFormation, achieving a 30% reduction in server costs through serverless architecture.
- **Optimized serverless infrastructure** and communication by enhancing DynamoDB with efficient indexing, adaptive capacity, and seamless Lambda integration to handle 10,000+ requests per second; integrated AWS SQS for reliable message queuing, asynchronous processing, and robust microservice fault tolerance.

Cooptem | LLM Research Intern

12/23 – 01/24

- **Enhanced generative AI model accuracy** with advanced Retrieval-Augmented Generation (RAG) techniques, integrating OpenAI API with dense vector retrieval using vector embeddings, Llama-Index for efficient indexing, which led to cut inference time by 40% and improving retrieval precision for better scalability.
- **Leveraged data for real-time streams**, supporting high-throughput processing using Python and Weaviate to enable rapid access to over 1 million data records with sub-second latency.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

10/21 – 06/25

Walchand College Of Engineering, Sangli | CGPA: 8.19

KEY PROJECTS

Microservices Mania

- **Engineered a resilient microservices architecture** using Spring Cloud with Eureka for service discovery, API Gateway, and Resilience4J for circuit breakers, achieving 99.9% uptime and fault-tolerant service operations; enhanced system reliability with distributed tracing via Spring Cloud Sleuth and Zipkin.
- **Streamlined microservice scalability** and security by implementing event-driven architecture with Apache Kafka to process millions of daily events, secure authentication using Keycloak, and containerized services with Docker; established robust monitoring with Prometheus and Grafana.

CSI-S3-Spec

- **Developed a CSI driver for Kubernetes** that seamlessly integrated S3 storage, allowing pods to mount S3 buckets as native filesystems which supported efficient data access and management across large-scale deployments and improved storage scalability for handling data workloads exceeding 100TB.
- **Configured a customized Kubernetes ecosystem** with tailored storage classes and integration with S3-compatible services, used shared mounts for Docker, implemented Kubernetes Secrets for securing sensitive information, and created automated deployment scripts streamlining the deployment process and increasing container performance.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++

Databases: Relational Databases, BigQuery, PostgreSQL, DynamoDB, MongoDB, Weaviate

Cloud & AWS Skills: AWS Lambda, S3, SQS, Bedrock, Cognito, SDK, CLI, GCP, Linode

Tools & Infrastructure: Kafka, Docker, Kubernetes, Apache Airflow, Linux, Git, Maven, CMake

Frameworks/Libraries: Spring Cloud, Django, FastAPI, Flask

COURSEWORK

Distributed Systems & Operating Systems • Database Systems & Architecture • Data Structures and Algorithms • Computer Networks • Cryptography and Network Security

ACHIEVEMENTS

- **Won Veritas Uconnect 2.0** by delivering a standout software engineering solution.
- **Runner-up at L&T India Createch Hackathon 2024** for an innovative software solution tackling a real-world problem in the Industry.
- **Secured first place at Neuro-Hack** by Whirlpool Corp, creating an AI Chatbot enabling users to talk with Data.
- **Reached the finals in Bosch Auto Vision X** for innovative work in automotive technology, including Image Binding using C++.
- **Advanced to the semi-finals of Flipkart GRID 5.0**, showcasing strong Computer Science fundamentals and delivering the complete project within a strict deadline.
- **Bestowed as Knight on Leetcode** (peak rating 2132) and **Rated 5 Star on Codechef** (peak rating 2012), solving over 1,500 coding problems to master complex algorithmic patterns and concurrency challenges.